Jayant Pratap Singh

AI-Native Full Stack Developer

Delhi, India | +91 7819016236 | ranajayant527@gmail.com | GitHub | LinkedIn

Summary

Backend-focused Full Stack Developer building secure internal web platforms, REST APIs, and database-driven applications with Node.js, TypeScript, React, and PostgreSQL. Experienced in authentication, role-based access control, digital-signature workflows, and event-driven systems using Kafka and RabbitMQ. Use AI coding assistants to speed up implementation, debug issues, and explore codebases, reviewing all output for correctness, maintainability, and security.

Professional Experience

Full Stack Developer October 2025 – Present
Nirnaya System (Army Software Development Center) *New Delhi, India*

- Develop secure internal web applications that digitize operational workflows, replacing manual, paper-based processes with centralized digital services.
- Build role-based dashboards and reporting views giving real-time visibility into workload, processing status, and operational trends based on user responsibilities and permissions.
- Design and maintain REST APIs and normalized PostgreSQL schemas for consistent data exchange, data integrity, and reduced duplication as the application grows.
- Improve responsiveness under load by optimizing database queries, pagination, and data-loading behavior.
- Implement authentication and role-based authorization (RBAC) to protect sensitive records and restrict actions by user permissions.
- Develop secure document workflows with controlled file access and server-side PDF watermarking for downloaded documents.
- Add input validation, payload protection, rate limiting, and secure session controls across application services.
- Use AI coding assistants to explore unfamiliar code, speed up implementation, and investigate defects; review all generated changes for correctness, maintainability, and security.

Full Stack Developer August 2025 – October 2025
UPSALA DEFSOL *New Delhi, India*

- Developed modular full-stack web applications with clear separation between frontend interfaces, backend services, and database layers.
- Created reusable application components to keep UI consistent and simplify ongoing development and maintenance.
- Designed and implemented REST APIs supporting full CRUD (Create, Read, Update, Delete) operations.
- Implemented authentication, authorization, and role-based access control (RBAC) to restrict features and data access by user role.
- Investigated and resolved production defects, including backend integration issues affecting communication between application components.
- Improved application reliability through stronger error handling and fixes for failures across frontend and backend workflows.
- Tested API endpoints with Postman to verify request handling, authentication, and role-based access before release.

Projects

ActionHub – Automation Platform | *Next.js, Node.js, Express.js, PostgreSQL, Kafka, Prisma*

- Built a full-stack workflow automation platform where users select triggers, add ordered actions, and publish trigger-action workflows, with a visual workflow builder and management dashboards.
- Developed Express REST APIs for registration, sign-in, JWT-authenticated workflow operations, and trigger/action catalog retrieval, with Zod request validation.
- Split system into microservices for API management, webhook ingestion, event processing, and action execution; ingestion service accepts external payloads and records workflow runs for asynchronous processing.
- Implemented the Outbox Pattern with a PostgreSQL outbox table, saving workflow-run records and pending events in one database transaction for reliable execution.
- Built a polling processor that publishes pending outbox events to Kafka, and a Kafka consumer worker that executes queued email actions via SMTP.
- Designed Prisma data models for users, workflows, triggers, ordered actions, workflow runs, and outbox events, with JSON fields for flexible metadata.

DocEngine – Distributed Document Processing Platform | *Java 21, Spring Boot, PostgreSQL, RabbitMQ, AWS S3*

- Built an asynchronous document-processing platform that converts uploaded CSV files into PDFs via background jobs, with Spring Boot REST APIs for upload, job submission, status polling, and API-key tenant authentication.
- Decoupled API requests from PDF generation using RabbitMQ and a dedicated worker service; enforced tenant-scoped job lookups so clients cannot access other tenants' jobs.
- Implemented idempotent job submission with tenant-scoped keys and request hashing to block duplicates and detect key reuse with different payloads.
- Built atomic job claiming so only one worker processes a job at a time, with processing leases and periodic lease renewal; expired leases let other workers reclaim stalled jobs, enabling worker recovery after crashes or interruptions without duplicate processing.
- Developed CSV parsing and PDF generation handling varying headers, quoted fields, and multiline values; stored files in AWS S3 and served downloads through short-lived presigned URLs.
- Containerized services with Docker Compose, deployed on AWS EC2, and added Prometheus and Grafana for monitoring.
- Used k6 to load-test CSV upload and job-submission endpoints at 100, 200, and 500 concurrent virtual users.

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, Java, SQL
Web Technologies: HTML5, CSS3
Frontend Development: React.js, Next.js, Redux, Tailwind CSS, ShadCN UI, React Router, Context API
Backend Development: Node.js, Express.js, Spring Boot, REST APIs, Webhooks
Authentication & Access Control: Authentication, Authorization, Role-Based Access Control (RBAC), JWT, Session Management, Rate Limiting
Databases & ORM: PostgreSQL, MongoDB, MySQL, Prisma
Messaging & Distributed Processing: Apache Kafka, RabbitMQ, Event-Driven Architecture, Asynchronous Processing, Outbox Pattern, Idempotency
Cloud & Deployment: AWS (EC2, S3), Docker, Microsoft IIS, Vercel, Netlify
Security Tools & Protocols: OWASP ZAP, Burp Suite, Nmap, HTTPS, SSL/TLS, Digital Signatures
Monitoring & Testing: Prometheus, Grafana, k6
AI Engineering: AI Agents, Generative AI, Prompt Engineering, AI Workflows
Developer Tools: Git, Postman, Visual Studio Code, AI Coding Assistants

Education

College of Engineering Roorkee — 2021 – 2025
Bachelor of Technology in Information Technology — *Roorkee, India*

Certifications & Achievements

- Completed the Full Stack Developer Course - 100xDevs.
- Completed the IBM Full-Stack JavaScript Developer Professional Certificate - Coursera.
- Completed the Advanced Artificial Intelligence and Machine Learning course - Koenig Solutions.